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Enhancing Mobile Network Performance through ORAN-Integrated UAV-based Mobility Management

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***Abstract*—The open radio access network (O-RAN) architecture promotes the evolution of the RAN through the disaggregation of functions, the adoption of open interfaces, and the instantiation of a hierarchical closed-loop control architecture managed by the RAN intelligent controllers (RICs). This architecture enables interoperability, flexibility, and intelligence between different network elements and the integration of various artificial intelligence (AI) and machine learning (ML) applications into the network. One of the key benefits of O-RAN is the continuous monitoring of quality-of-service (QoS) metrics and network resources, which can be used to make optimized decisions about resource allocation, management, and scaling. One of the network features that can benefit from this approach is the use of unmanned aerial vehicles as flying base stations (UAV-BSs), which can be deployed to provide wireless connectivity to ground devices, complementing the terrestrial infrastructure, such as in high-demand scenarios, infrastructure failures, or disasters. However, deploying and controlling UAV-BSs are complex challenges involving aspects such as positioning, power consumption, interference, safety, and mobility. In this context, this paper proposes an approacH that Adjusts drones as flying Base statIons using weighTed k-meAns for the oraN archiTecture (HABITANT), a new approach for the O-RANs architecture, consisting of the use of a positioning algorithm running at an extensible application (xApp) to calculate optimized locations for UAV-BSs. The algorithm takes into account the QoS metrics of user applications, as well as the physical and operational constraints of UAV-BSs. The goal is to provide a more efficient and adaptive solution to the UAV-BS positioning problem, improving the network coverage and the QoS of the users.**

***Index Terms*—UAV, UAV-BS, 5G, Open-RAN, RIC, flight-assisted networks**

I. Introduction

Open radio access network (O-RAN) and its embodiment through the open-RAN alliance (O-RAN alliance) specifications have the potential to transform the telecommunication ecosystem. O-RAN promotes virtualized and disaggregated RANs, where disaggregated components are connected via open interfaces and optimized by RAN intelligent controllers [1]. The result is a new RAN design, deployment, and operations paradigm, where O-RAN can be built with multi-vendor, interoperable components, and programmatically optimized through a centralized abstraction layer and data-driven closed-loop control [2].

The O-RAN architecture is especially relevant for fifth generation (5G) networks and beyond, which is characterized by various options for serving users through the use of radio access technologies (RATs), frequency bands, or cells, and diversified mobile traffic of heterogeneous QoS demands. The challenge for network operators is to effectively decide on how to serve a particular user (or a group of users) to meet their per-slice or per-QoS type demands. This is important to avoid coverage problems and load unbalancing between cells, e.g., in heterogeneous network (Het-Net) scenario situations, where a particular cell is highly loaded, and a neighboring cell serves only a few users [3].

O-RAN enables the collection and analysis of different QoS metrics from different network elements, such as, radio units (RUs), distributed units (DUs), centralized units (CUs), and RICs. In this way, O-RAN can be used for overall QoS monitoring, including signal strength, signal to interference plus noise ratio (SINR), block error rate (BLER), throughput, latency, and jitter. The network management services can benefit from the near-real-time measurement, and use, for instance, AI and ML techniques to optimize the network performance. For example, in a network assistance scenario, a mechanism can collect QoS performance metrics and deploy UAV-BSs to enhance the overall QoS performance of user equipment (UE) on the ground [4]. Another algorithm can perform load balancing, handover, or cell reconfiguration to enhance the QoS. A third system can use the QoS parameters to provide efficient resource provisioning to users requesting high-resolution video services [5], or use reinforcement learning to dynamically adapt the network parameters to changing traffic patterns and user demands [6].

The near-RT RIC is a critical component of the O-RAN architecture that directly affects the monitoring feature. It is a platform that provides RAN control and optimization functions by interacting with the eNodeBs (eNBs) or next generation nodeBs (gNBs) via the E2 interface. The near-RT RIC hosts applications, called extensible applications (xApps), which are

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the core of the decision-making process and responsible to run the algorithms behind the network management services (e.g. QoS monitoring and optimization, traffic steering, load balancing, interference management), using the data, and services that the near-RT RIC provides [7].

The E2 service model is a specification that defines the data structures and services used by the E2 interface for a particular use case or scenario. The O-RAN alliance has defined several E2 service models for different purposes, such as key performance metric (KPM), RAN control (RC), and network interface (NI). Each E2 service model defines a set of indicators, parameters, actions, and relevant policies for the use case or scenario. For example, the KPM service model defines how to measure and report various performance metrics from the RAN to the near-RT RIC, such as cell load, user throughput, and latency. The RC service model defines how to control and optimize various aspects of the RAN, such as traffic steering, load balancing, and interference management.

In this paper, we explore a use case scenario in the context of the use of the O-RAN architecture, which is an online mobility management system of UAV-BSs that provides network coverage and QoS support to mobile UEs on the ground space. The proposed approach, denoted as HABITANT, aims to optimize the UAV-BS mobility management and the network coverage by considering the UE's mobility patterns and the terrestrial UE's network traffic demand. It utilizes the network parameters gathered from the RAN, such as the user throughput, to inform decisions regarding the mobility management service of UAV-BSs. This approach can be implemented as an xApp hosted by the near-RT RIC, which can interact with the UAV-BS and the terrestrial UE via the E2 interface. Additionally, this work explores the trade-off between the frequency of E2 control messages and the accuracy of positioning determined by the xApp.

The rest of the paper is organized as follows: Section II presents a set of works that consider either the use of UAV-BS or the O-RAN architecture for network management. Section III describes the proposed system along with the O-RAN architectural features, the employment of QoS parameters, the positioning system approach, and the management of the control layer packet transmissions. The details about the scenario design and simulation environment are described in Section IV along with the main findings. Finally, Section V shows the open issues that future works can address, and concludes the main discussed ideas in this paper.

II. Related works

Bertizzolo et al. [8] propose a solution to support unmanned aerial vehicle (UAV)-based video streaming applications on 5G cellular networks by using O-RAN. The solution considers a UAV as a UE, and consists of a closed-loop control system that optimizes the UAV's location and transmission direction based on the O-RAN architecture, as well as a learning approach that combines multi-agent reinforcement learning and online learning methods to solve the optimization problem. The goal is to improve the overall network performance of the ground UEs by adapting the UAVs' positions and antenna direction. Performance evaluation experiments were carried on a real-world RAN test-bed and large-scale simulations, and the results show that it outperforms traditional solutions regarding resource utilization, utility values, and service acceptance rate.

Polese et al. [9] mentions that O-RAN can support, through xApp, the UAV position prediction for handover improvement. In this case, the UAV is considered a UE. However, our approach extends this concept by considering UAVs as eNBs (UAV-BSs) rather than traditional UEs. Our goal is to leverage O-RAN architecture, not to predict UAV positions for handover improvements, but to optimize UAV-BS positions strategically and QoS support for the devices on the ground space. By positioning UAV-BSs, we aim to enhance the availability of communication resources by situating them in proximity to high-demanding areas. Additionally, our scenario considers the specific types of traffic that this set of UAV-BSs will manage, thus enabling a more targeted network resource provisioning for UEs with different traffic requirements.

Pham et al. [10] introduces a framework to integrate multi-UAV technology seamlessly with the RIC to enhance the network performance and operational efficiency. It introduces a learning solution for the flying O-RAN architecture to optimize UAV-BS trajectory and distributed resource units allocation. Besides, it evolves the scenario presented developed by Pham et al. [11] and propose a joint optimization problem of multi-UAV trajectory and offloading tasks. The prior work considers a wireless sensor network (WSN) scenario, where the UAV-BSs offload tasks from static sensors, while the later considers mobile users that each require a single task. However, it does not consider the impact of different types of users or applications on the optimization objective and constraints. For example, some users or applications requirements may vary along the time upon changes on mobility, interference, or coverage (from near base stations (BSs)). The users' application may also have different QoS requirements, data rates, or security levels than others. This scenario could require a more user-centric and application-aware optimization approach that can prioritize and satisfy different users' or applications' needs and preferences, and the trade-offs among different QoS metrics.

In summary, our HABITANT considers the use of the O-RAN architecture to improve UAV-BSs assisted network management through the use of multiple UAVs acting as UAV-BSs. It considers the presence of mobile UEs, each running applications with different throughput requirements. Also, it extends the utilization of the O-RAN's control messages, namely throughput field, enabling a more accurate representation of the network and communication conditions of each UE and the different services being executed. In addition, we introduce an approach to manage the control messages that significantly reduce the control layer overhead without compromising: (1) the application of the UEs; (2) the accuracy of the positioning algorithm; nor (3) the performance of the xApp running the UAV-BS position optimization algorithm.

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III. HABITANT

This section provides an overview of the O-RAN architecture, E2 interface, and service models defining HABITANT. It also provides details on the developed xApp that computes optimized positions for UAV-BSs using the QoS and position coordinates received from the RIC component through the E2 interface, and the mechanism to avoid redundant control packet transmission without reducing the service performance.

A. Architecture and QoS update

Our system approach, HABITANT, takes advantage of the O-RAN architecture to provide flexibility and interoperability thanks to the hardware and software components decoupling present in RAN. The QoS parameters, such as throughput, offer valuable information about the UE's application and network conditions of the UEs, enabling the RIC component to make customized decisions.

The O-RAN E2 interface plays a critical role in our system by enabling the control and management functionalities for the RU, allowing the near-RT RIC to manage, configure, monitor, and maintain RU operations. The UAV-BS positioning algorithm gathers key information (position and QoS) from the UEs and sends it to the near-RT RIC, where a positioning algorithm defines the new positions for the existing UAV-BSs. The QoS metrics are crucial indicators for network operators, application developers, and users to evaluate the network performance. Figure 1 presents the part of the O-RAN architecture where the QoS information is exchanged.

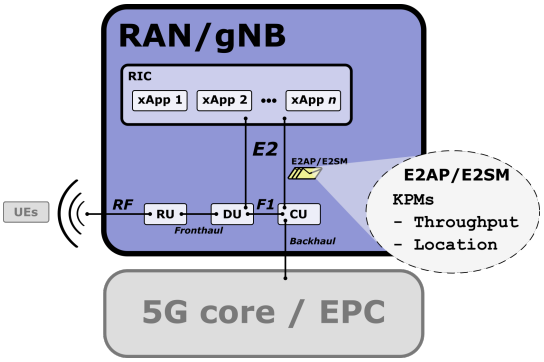

Fig. 1: RAN and network architectural and components.

It is important to notice that the development of an xApp does not necessarily depend on the technology it will use, whether fourth generation (4G) or 5G. However, it depends on the interface and service models defined by the O-RAN alliance for communication between the near-RT RIC and the eNBs or gNBs. These models specify the data structures, messages, procedures, and functions supported by the E2 interface for different RATs – long term evolution advanced (LTE-A) and fifth generation new radio (5G-NR). The O-RAN architecture does not directly provide interoperability between LTE-A and 5G-NR. Instead, it provides a common framework for developing and deploying xApps that can work with both RATs, as long as they comply with the O-RAN specifications for the E2 interface and service models [1, 2].

B. O-RAN xApps

In the O-RAN architecture, xApps are essential applications that operate on the near-RT RIC. These xApps consist of multiple micro-services that receive input data from interfaces between the RIC and RAN functionality. They then provide additional functionality as output to RAN, utilizing AI/ML algorithms for near-RT control.

An xApp is a piece of software application running on the near-RT RIC. It can be used to various purposes, including resource allocation, connection management, and frequency scanning, enabling intelligent and automated control of the RAN. Thus, it is a crucial component of the O-RAN architecture. Additionally, the RIC serves as a platform for designing and deploying these network controllers and applications.

Our developed xApp computes the optimized positions for the UAV-BSs. It utilizes the throughput metric and position coordinates received from the RIC component through the E2 interface, and determines an optimized set of positions for the UAV-BS. The QoS is constantly received at the near-RT RIC from the DU/CU modules at the UAV-BS, merged into JavaScript object notation (JSON) format, and then sent through the E2 interface from a packet data convergence protocol (PDCP) exit point in the CU. The JSON format is defined by O-RAN alliance [12] and contemplates the throughput field to be filled with the respective information, becoming part of the QoS information. Once this information is received by the positioning xApp, a new set of positions is computed and then sent back to each UAV-BS, which moves to the appropriate position.

The positioning of UEs in the scenario can be estimated by eNB in different ways, such as uplink time difference of arrival (UTDoA), observed time difference of arrival (OTDoA) or global navigation satellite system (GNSS) and their variants [13, 14, 15]. The O-RAN architecture foresees the use of these techniques. It provides the necessary fields, such as the angle-of-arrival information.

C. System design and flow

The HABITANT starts by gathering all the UEs' positions and current experienced throughput. These metrics are then used as a weighted score that will encourage the solution to prioritize the worst attended and high demanding set of UEs. The score range is normalized, and the correspondent integer value is calculated respecting the weights' proportionality. At the end, the UEs' positions are duplicated according to the ratio defined by the weight values. Thus, the higher the weight of a certain UE, the higher the number of duplicates of its position in the new UE position list $\mathbf{P_{UEs_w}}$. Algorithm 1 shows the calculation of $\mathbf{P_{UEs_w}}$.

The locations of the UAV-BSs are determined by the centroids of a weighted list of positions that was previously calculated. To find these centroids, the K-Means algorithm is utilized because of its speed and efficiency. It splits the dataset into K clusters by minimizing the sum of squared distances between data points and their respective cluster centroids. In our case, the algorithm uses the Euclidean distance as a

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Algorithm 1: Weighted UEs' position list.

Input: N_{UEs}, $\mathbf{P_{UEs}}$, $\mathbf{Th_{UEs}}$
Output: $\mathbf{P_{UEs_w}}$
$\mathbf{W} = (w_1, w_2, \ldots, w_{N_{UEs}})$;
for i = *0* **to** $\mathbf{P_{UEs}}$ **do**
  $\mathbf{W}$[i] = $(\mathbf{Th_{UEs}}[\mathrm{i}])^{-1}$;
end
$\mathbf{W} = \mathbf{W} \times \max(\mathbf{W})^{-1}$;
$\mathbf{W} = \mathbf{W} \times \left(\max(\mathbf{W}) \times \min(\mathbf{W})^{-1}\right)$;
$\mathbf{P_{UEs_w}}$ [3][N_{UEs} + $\sum_{i=0}^{n-1} \lfloor \mathbf{W}[\mathrm{i}] \rfloor$]=
$\left[0\ 0\ \ldots\ 0\quad 0\ 0\ \ldots\ 0\quad 0\ 0\ \ldots\ 0\right]$;
index = 0;
for i = *0* **to** $\mathbf{P_{UEs}}$ **do**
  $\mathbf{P_{UEs_w}}$[0][index] = $\mathbf{P_{UEs}}$[0][i]; $\mathbf{P_{UEs_w}}$[1][index] = $\mathbf{P_{UEs}}$[1][i]; $\mathbf{P_{UEs_w}}$[2][index] = $\mathbf{P_{UEs}}$[2][i]; index ++;
  for j = *0* **to** $\mathbf{W}$ **do**
    $\mathbf{P_{UEs_w}}$[0][index] = $\mathbf{P_{UEs}}$[0][i];
    $\mathbf{P_{UEs_w}}$[1][index] = $\mathbf{P_{UEs}}$[1][i];
    $\mathbf{P_{UEs_w}}$[2][index] = $\mathbf{P_{UEs}}$[2][i]; index ++;
  end
end

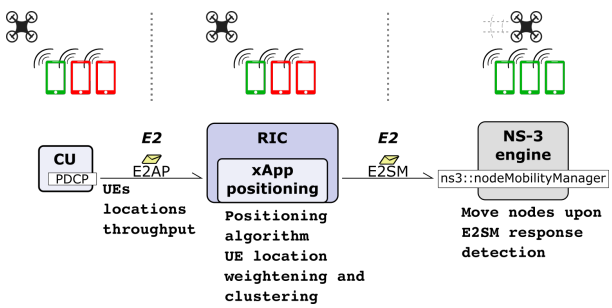

Fig. 2: Flow of the message update containing the UEs' QoS information.

distance measure to assign each point to its nearest group. Then, it calculates the average of the points within each group to update the centroids. The number of centroids is determined based on the cost-effective measurement index defined in [4]. Figure 2 illustrates the flow mechanism that enables the UAV-BSs adaptive positioning. The E2 control messages travel from the collecting point at an eNB, and the calculated positions are managed by the correspondent UAV-BSs mobility module from the simulator.

The allocation of new positions to UAV-BSs in dynamic and mobile scenarios can be challenging. To address this, the Hungarian algorithm is used to assign new positions to UAV-BSs based on their current locations. Originally developed for optimal worker-to-task assignment, the Hungarian algorithm [16] minimizes the total displacement of UAV-BSs by finding a minimum number of lines that cover all zero entries in a cost matrix. The distances to the new set of positions form the cost matrix, allowing for optimal assignment while minimizing unnecessary or costly displacements. This algorithm is especially useful for UAV-assisted networks, minimizing displacement costs.

Figure 3 shows one of the samples of the UAV-BSs adjustment when receiving a new set of positions defined by the xApp from HABITANT. In this case, the UAV-BSs take around nearly 14 seconds to move from their current locations towards the xApp positions, getting closer each second. It starts by showing the moment of greatest difference between the UAV-BSs positions and the new positions densities, on the left. After the assignment procedure, each UAV-BS moves towards its assigned xApp position. The last image on the right shows the moment when the UAV-BSs arrive at their new locations, being possible to observe a more significant similarity between the location density of the UAV-BSs and the newly calculated positions.

D. Control layer and system management

The O-RAN architecture requires the exchange of control messages between the eNBs (macro cell (MC), small cell (SC) or UAV-BS) and the RIC, the latter being responsible for the decision-making process. These control messages record each UE's throughput and positioning, essential for improving decision-making in the O-RAN architecture by the xApp.

Generating and using these control messages is as follows: each eNB receives data packets from the connected UEs. The quantity and frequency of these data packets depend on the application running in each UE, such as augmented reality (AR), virtual reality (VR), or video over demand (VoD). Upon receiving and recognizing these data packets, the eNB accounts for the QoS of each UE. Subsequently, the eNB creates a control packet containing this information and sends it to the RIC.

Managing these control messages involves determining the frequency of creating and sending these control packets to the RIC from the eNB. Over time, the number of control packets influences the accuracy of the QoS information. This work implements a method to aggregate QoS values within a time interval to accurately reflect the state of each UE.

Our system aims to maintain a high QoS for an extended period. However, constantly sending the QoS can cause information arriving at the RIC to become redundant. It can also potentially cause bandwidth overload. Sending this information after a long time frame may result in outdated data and inaccurate UAV-BS positioning. To address this, we reduce the number of control packets generated by the UE without causing any loss of information or degradation of service from the UE's perspective.

Control packets are generated upon reception of a data packet sent by each UE. The QoS metrics of the received data packet are inserted into this control packet. Each control packet has a field designated for the value of throughput and coordinates of a single UE. Such control packets are then sent to the xApp through the E2 interface, so that adjusting the positioning of UAV-BSs considers such metrics.

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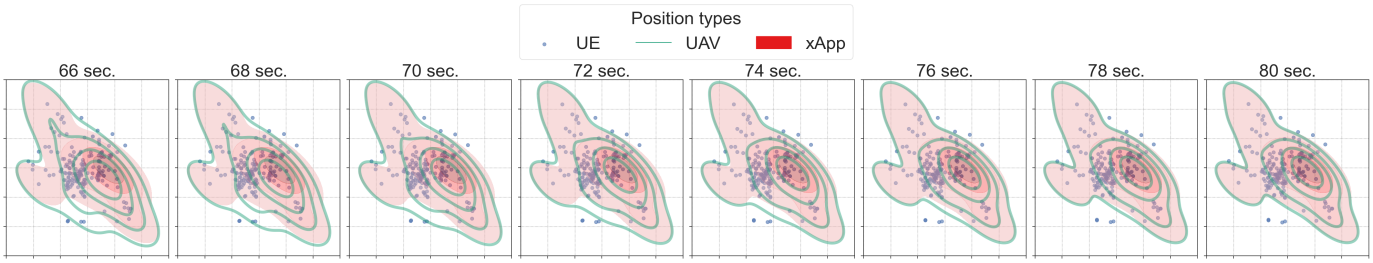

Fig. 3: Density similarity over time among UEs', UAVs' and xApp's positions.

However, the distance covered by the set of UEs in a short interval of time, for example, 1 second, does not justify the immediate calculation of the UAV-BSs' positions. Such devices can provide reasonable connectivity in an area of 200 meters range, and the creation of control packets is constant as long as the UE application is running. This results in a redundancy of information to be computed. To reduce the number of control packets, we aggregate the QoS values for the packets of each UE and send an average of these values within a fixed interval of 10 seconds. This value was defined arbitrarily upon the tests carried out. To understand the impact of managing the control packets transmission, we have compared to other two implemented approaches. The first one does no consider any management, so the control data is sent upon every received data packet. The second aggregates the information in an interval of 10 ms.

It is important to highlight that more detailed studies are necessary to obtain an optimized value related to the mobility characteristics of the devices such as the UEs. Within the current control packet management, we can attest to better bandwidth usage while keeping or, in some cases, improving UAV-BS positioning service performance. More details about the scenario as well as the performance of each of these system features are presented in the following section.

IV. Evaluation

This section presents the behavior and impact of the HABITANT approach in providing QoS support for a set of mobile UEs during networking congestion periods. We analyse the presence of UAV-BS and the impact of the O-RAN architecture under the approach introduced by HABITANT, and discuss the simulation results.

A. Methodology

This work establishes its foundational reference in the proposal context as the work presented in [4], but now extended with O-RAN and mobility management approach. We conduct simulation experiments to assess the effectiveness of our solution in flight-assisted networks. We use version 3.38 of the network simulator 3 (NS-3) simulator and the ns3-ORAN module [17] to perform this analysis. The ns3-ORAN module is a simple implementation of the O-RAN architecture that prioritizes ease-of-use and development speed of xApps, as well as the integration with the NS-3 simulator. It is equipped with enough functionality to create additional E2 service models for KPMs and RCs.

The experiments are conducted within a realistic scenario consisting of an urban area spanning 6 square kilometers, using the mobility pattern provided by the Cologne, Germany trace data set [18]. Additionally, the scenario incorporates four MCs and eight SCs, each equipped with multiple-access edge computing (MEC) servers that facilitate various edge services such as caching, encoding, and transcoding of multimedia content. The mobile UEs start connected to the closest MC or SC. Soon afterwards, the UEs randomly request one of the services defined in Table I, AR, VR or VoD, each associated with a specific throughput. The A2-A4-RSRQ handover procedure allows each UE to switch among the existing eNBs (MCs, SCs and UAV-BSs) based on its connected and the neighbors eNBs's reference signal received quality (RSRQ).

TABLE I: Services requirements

Service name	Throughput
AR	97 Mbps
VR	25 Mbps
VoD	10 Mbps

Moreover, the scenario incorporates 20 UAV-BSs that can be dynamically positioned to cater to user demands thanks to the positioning algorithm available in the xApp that runs at the near-RT RIC level. These UAV-BSs possess a maximum communication range of 250 m, a transmit power of 23 dBm, and a fixed altitude of 10 m – UAV-BS under the low altitude platform (LAP) group [19, 20]. The LTE-A and hybrid-building propagation loss models are employed to determine the transmission parameters. The simulation runs for 300 seconds and is repeated 20 times to ensure reliable results with a 95% confidence interval. The simulation parameters are presented in Table II.

B. Results

We first analyze the use case where a set of multiple UAVs acting as UAV-BS need to be managed, with the help of the O-RAN architecture. The objective is to cater to the UEs' needs, by strategically allocating them in a clustered fashion based on their positions and QoS metrics: we also consider a similar approach to update the UAV-BSs positions, this time following the algorithm running at the xApp.

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TABLE II: Simulation parameters for O-RAN UAV-BS deployment evaluation.

Parameter	Value
UEs mobility trace	Cologne city, Germany [18]
Scenario area	6 km^2 (3000m X 2000m)
Number of UEs	300
Number of UAV-BSs	20
Simulation time	300 seconds
Simulation runs for each scenario	20
Propagation loss model	hybrid buildings
LoS[1] to NLoS[2] threshold	250 m
eNB type of transmission	ITU's LoS
MC transmission power	46 dBm
UAV-BS, SC transmission power	23 dBm
MC and SC altitude	40 m
UAV-BS altitude	LAP (10 m)
UAV-BS maximum speed	30 m/s

[1] Line-of-sight.
[2] Non-line-of-sight.

As observed in Figures 4 and 5, having UAV-BS placed within a dynamic environment has a positive impact. The existing UEs take advantage of the communication resources provided by these UAV-BSs. They show that HABITANT improve the packet delivery ratio (PDR) and throughput rate by approximately 21%. The throughput rate achieved is the amount over and above the throughput required by the UE application.

Poorly assisted rate (PAR) is a metric used to measure the percentage of users who do not receive adequate flow for their applications. The flow requirement for each application varies based on the type and the QoS provided. To calculate the PAR, a flow threshold is arbitrarily defined for each application, equivalent to 80% of the required application throughput, as defined in Table I. In other words, PAR represents the proportion of users whose received throughput falls below 80% of its running application. The number of UEs who receive a throughput less than or equal to this threshold is then counted, and this number is divided by the total number of users.

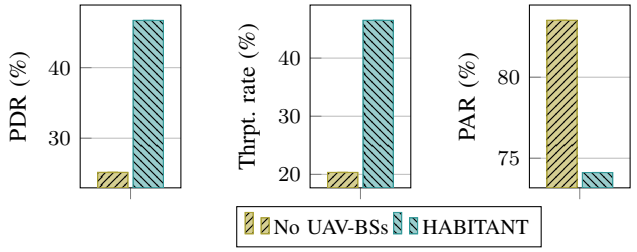

Fig. 4: Average QoS metrics performance of HABITANT against the scenario without UAV-BSs.

Within a dynamic scenario characterized by the mobility of UEs, the resource demands evolve geographically. The evolution is closely intertwined with the changing positions of UEs over time. This unique scenario aspect introduces a novel challenge, contrasting with the static scenarios discussed in [4]. It now needs that UAV-BSs dynamically adapt to the evolving landscape. These UAV-BSs can shift closer or farther away from regions with varying QoS, influencing network performance outcomes. HABITANT tries to provide the ideal positions (generated by the xApp) and the best combination between them and the UAV-BSs' current positions. Based on the conducted tests considering the parameter detailed in Table II, it becomes evident in Figure 5 that the movement of UAV-BSs significantly impacts network performance, especially during the first 20 seconds, when the network starts to become aware of the UEs QoS performance. Figure 5 depicts the performance over time, showing that HABITANT maintains the overall performance even with the UEs constantly moving.

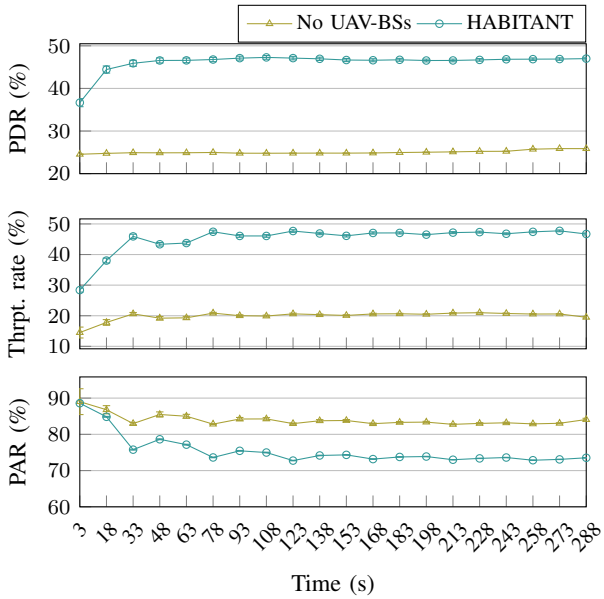

Fig. 5: The mean value of the overall impact of the UAV-BS presence over time.

Further tests were conducted to analyze the impact of managing the amount of E2 control messages sent by UAV-BSs, MCs and SCs thoroughly. The instantaneous transmission of control packets upon receiving a data packet prevents obtaining the most confident information from each UE. Furthermore, it is uncertain whether the O-RAN system can sustain higher volumes of control packets with given applications with much higher throughput. Thus, in the scenario of full control, the volume of control messages is reduced due to an underlying intelligence that consolidates possibly repetitive data, such as the position and QoS of each UE, within a specified interval. It is probable that, within this time-window, the QoS and position of a group of UEs remain relatively stable, thereby eliminating the need to generate a new set of positions for the surrounding UAV-BSs. In other words, full control consists in aggregating UE information over 10 seconds, allowing for more explicit photography of the regions with higher network resource demands. In Figure 6 it is possible to observe that, despite a decreased number of control messages (attributed

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to information aggregation), the user application maintains a similar performance level. In the absence of this management (no control), a substantial volume of control packets is transmitted by each existing eNB (MC, SC or UAV-BS). These packets may not necessarily carry the most valuable knowledge regarding the regions with higher network demand due to the high frequency of E2 control packets transmission. In partial control, the E2 control packets are aggregated for a brief interval (10 ms) before effective transmission.

Figure 6 also provides a significant observation: in the absence of this management (No control), a substantial volume of control packets is transmitted by each existing eNB (MC, SC or UAV-BS). These packets may not necessarily carry the most accurate knowledge regarding the regions with higher network demand. In partial control, the control packets are temporarily held for a brief interval (10 ms). Full control involves aggregating UE information over 10 seconds, allowing for more explicit photography of the regions with higher network resource demands. Notice that the objective of the full control is to decrease the amount of control messages, since there is intelligence to aggregate redundant information such as the position and QoS of each user equipment in the 10 seconds interval. The specific time required to update the UEs' QoS and position remains undetermined and will be explored in future works, potentially involving prediction and an analysis of UE mobility.

Even in partial control, a relatively high volume of control packets transmitted by the eNBs is still noticeable. This significant volume impacts the available communication resources, posing a significant risk of degrading service [21, 22]. Figure 6 shows the impact on the service provided by the UAV-BSs. With full control over the control packets, the service demonstrates enhanced performance regarding PDR, throughput rate, and the number of poorly served UEs.

It is important to emphasize that the aggregation interval directly correlates with the dynamic nature of the scenario. The aggregating interval is indirectly determined by UEs through their QoS metrics, identifying regions or locations with heightened network resource demands. Another consideration is the time the set of UAV-BSs takes to reach the locations specified by the xApp, which varies depending on how far the UAV-BSs are from these locations.

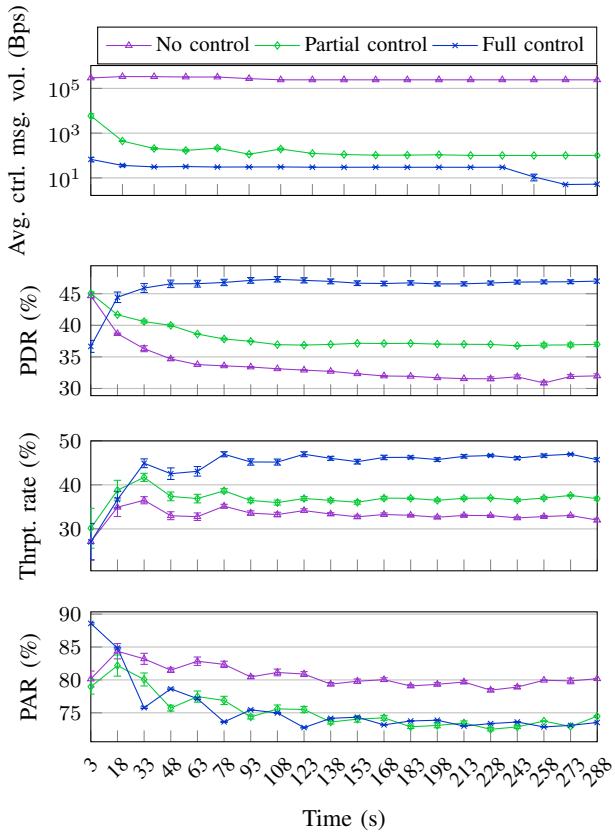

Fig. 6: Different management approaches for the control messages from the E2 interface.

V. Conclusion and Future Works

This paper discussed the employment of the O-RAN architecture, its features and impacts when considering UAVs as flying base stations to provide network assistance to mobile users. The proposed HABITANT is a continuing evolving approach that aims to optimize the mobility management of UAVs and the network coverage by considering live UEs' mobility patterns and their traffic demands. HABITANT includes an xApp hosted by the near-RT RIC, which can interact with the UAV-BSs. It improves the network performance by using the data collected through the O-RAN E2 interface, such as user throughput and number of properly assisted UEs, and make smart decisions on the UAV-BSs' mobility management service. It also carefully manages the number of control packets avoiding network overheads and keeping a reasonable UEs' application performance.

Future work is being developed in various angles, such as developing appropriate handover guidance mechanisms for UEs to select nearby idle UAV-BSs resources. By leveraging machine learning and predictive analytics, it would be possible to anticipate the long-term and short-term movement of UEs and proactively position and adjust UAV-BS to ensure continuous and enhanced resource provisioning and coverage, thereby diminishing mobility-related energy expenditure of the UAVs. Additionally, future research could explore how flight controllers can effectively manage the feasibility of reaching ideal locations (the ones defined by the xApps) and to what extent it affects the network performance. To ensure that the network can meet the dynamic demands of users without compromising the physical integrity of UAVs, it is also important to separate the positioning calculation procedure, done at the near-RT RIC level, from the flight controlling procedure, done at the UAV mechanical level, considering the impact of various flight control parameters, such as speed, altitude, collision avoidance triggers, and trajectory planning. The analysis of diverse regulatory frameworks governing the utilization of UAV-BSs encompasses a wide-ranging domain that significantly influences the adoption of this technology

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and necessitates, thorough future investigation, to ensure safety and performance.

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